



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 87



KAIAO CENTRIFUGAL PUMPSET
Model : KDP40

KAIAO KDP40 Self-priming Centrifugal Pumpset

Test Report No.	:	2013 - 87
Date of Test	:	August 9, 2013
Test Valid Until	:	August 9, 2018
Brand	:	KAIAO
Model	:	KDP40
Serial No.	:	No data
Type and Size	:	Single-end suction, 100 x 100 mm, Self-Priming Centrifugal pumpset
Manufacturer	:	WUXI KAIAO POWER MACHINERY CO., LTD. Wuxi, Jiangsu Province, China
Test Requested by	:	ALL CERTIFIED EQUIPMENT TRADING CORPORATION 203 APC Building, 1186 Quezon Avenue, Quezon City
Machine Classification	:	Commercial

SUMMARY

The KAIAO KDP40 Self-priming centrifugal pumpset was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump were mounted on a steel frame.

The Maximum Total Head (H) was 27.6 m at zero discharge condition. The shaft speed of the pumpset at this point was 3747 rpm, with Engine Fuel Consumption Rate of 1.51 L/h. The Maximum Discharge Capacity of 19.7 L/s occurred at 6.61 m Total Head. At this point the Maximum Fuel Consumption Rate of 1.68 L/h at this point. The shaft speed at this point was 3691 rpm. The Maximum System Efficiency was 11.6% occurred at 14.0 L/s Discharge Capacity and 14.4 m Total Head.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.05 m. The Net Positive Suction Head Available at this point was 1.81 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pumpset was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on August 9, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. KAIAO KDP40 Pumpset on the test set-up

METHOD OF TEST

The pump was tested following the “ Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The KAIAO KDP40 Self-priming Centrifugal pumpset shown in **Figure 2**, was a single-end suction, 100 x 100 mm, centrifugal pump closed-coupled to a 6.60 kW (8.84 hp) KAIAO diesel engine, Model: KA186F. The pumpset was mounted on a steel base. The suction inlet and discharge outlet were threaded flanges and designed for a 100 mm diameter pipe. The impeller was a semi-open type made of cast iron material connected to the pump shaft through a thread. It was housed on a cast iron volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. The suction inlet was provided with a swing type one-way valve to retain the sufficient amount of liquid inside the chamber necessary to prime the pump. The discharge outlet was located on top of the pump facing upward. The stuffing box used a mechanical seal.

The primemover, a KAIAO Diesel engine, Model: KA186F was a four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with fuel tank located on top of the engine. It utilized a dry type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started using a rope-recoil system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

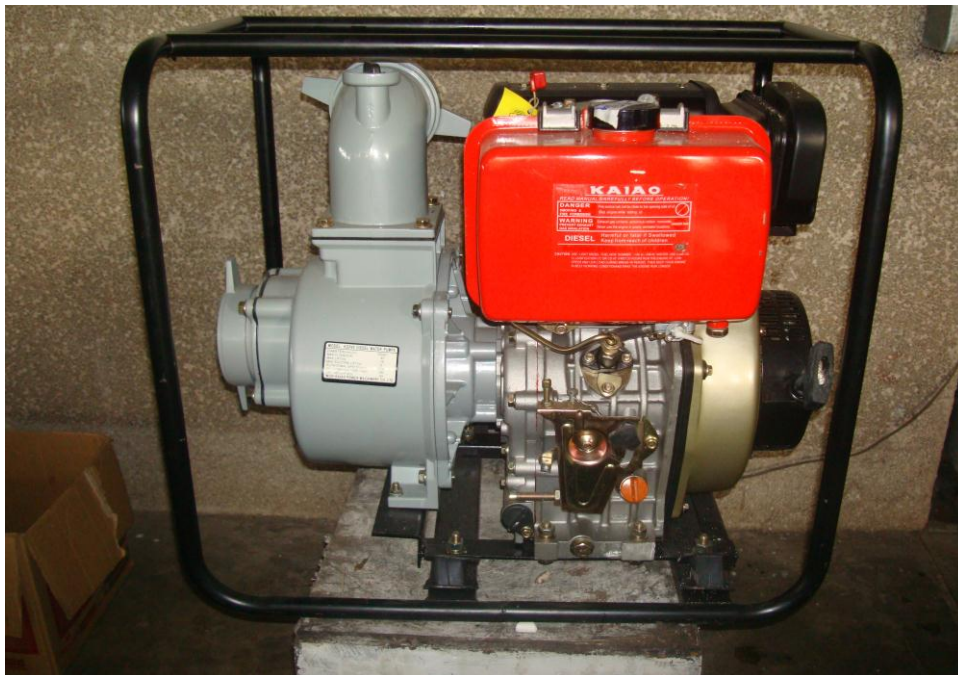


Figure 2. KAIAO KDP40 Self-priming Centrifugal Pumpset

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KAIAO KDP40 Self-priming Centrifugal Pumpset

1.	Brand	:	KAIAO
2.	Model	:	KDP40
3.	Serial No.	:	No data
4.	Make	:	WUXI KAIAO POWER MACHINERY CO., LTD., Wuxi, China
5.	Type	:	Single-end suction, Self- priming centrifugal pumpset
6.	Size (mm)	:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	6440
7.2	Width (mm)	:	472
7.3	Height (mm)	:	600
7.4	Weight (kg)	:	58.0
8.	Impeller		
8.1	Type	:	Semi-open
8.2	Diameter (mm)	:	128
8.3	Width (mm)	:	27
8.4	No. of vanes	:	3
8.5	Material	:	Cast iron
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	100
9.3	Material	:	Aluminum

Table 1. (Cont'n)

10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	100
10.3	Material	:	Aluminum
11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12.	Stuffing box	:	Mechanical seal
13.	Rotation	:	Counter clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	3600
15.	Rated Maximum Discharge Capacity (L/s)	:	11.1
16.	Rated Maximum Total Head (m)	:	16
17.	Primemover (Based on nameplate)		
17.1	Brand	:	KAIAO
17.2	Model	:	KA186F
17.3	Serial No.	:	no data
17.4	Make	:	WUXI KAIAO POWER MACHINERY CO., LTD., China
17.5	Type	:	Four stroke, air-cooled single vertical cylinder diesel engine

Table 1. (Cont'n)

17.6	Rated Output Power (kW)		
	Maximum	:	6.60 @ 3600 rpm
	Continuous	:	6.30 @ 3600 rpm
17.8	Bore x Stroke	:	86 x 72 mm
17.9	Displacement Volume (cc)	:	418
17.10	Cooling System	:	Air-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Dry type (cylindrical pleated paper element)
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : KAIAO Centrifugal Pumpset
Model : KDP40

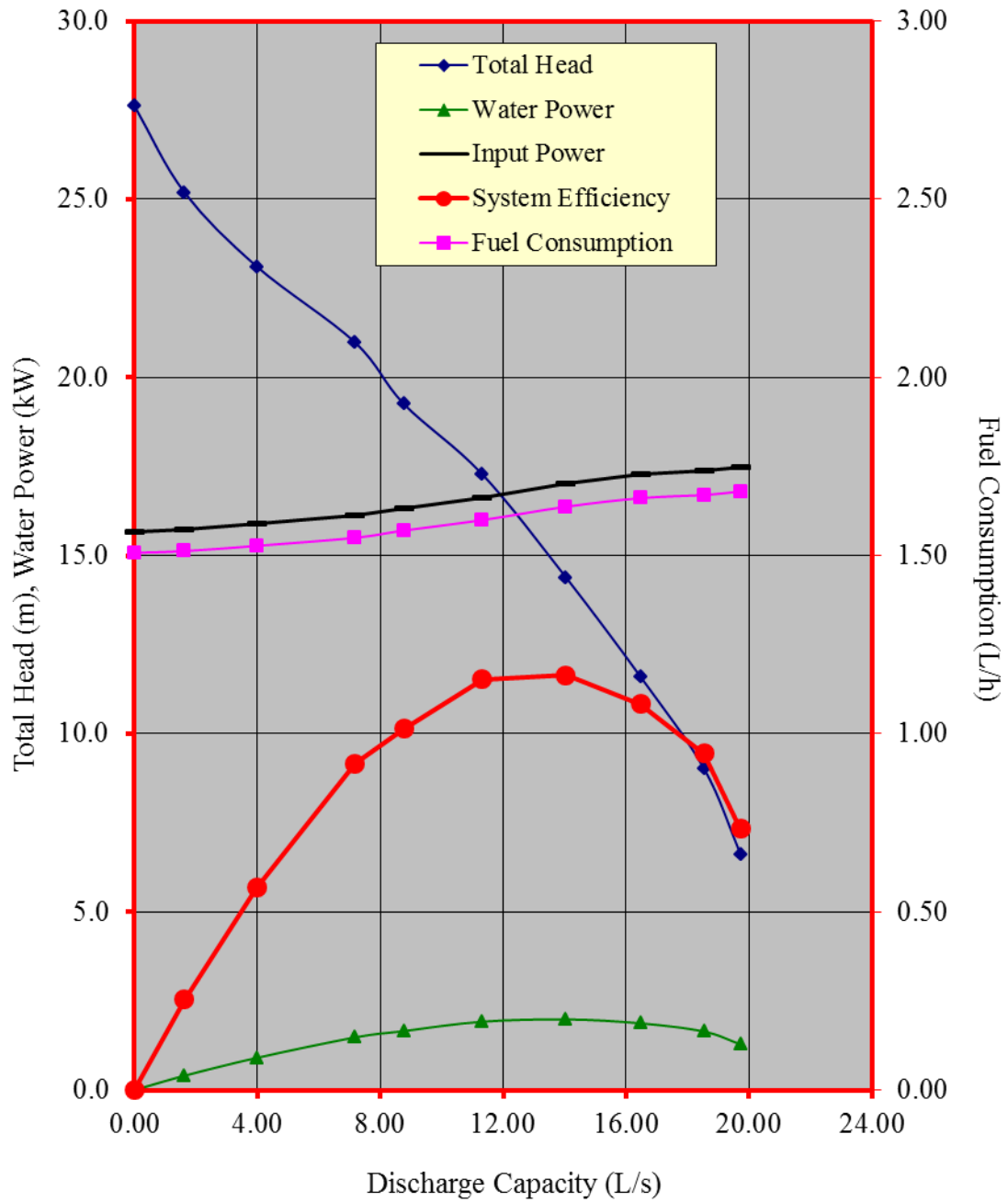
Date of Test : August 9, 2013

Test Condition

Ambient Temperature (°C) : 33.7
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 28.0

Table 2. Results of Performance Test

PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY *%)
3747	1.51	27.6	0	0	15.7	0
3732	1.51	25.2	1.62	0.40	15.7	2.54
3723	1.53	23.1	3.98	0.90	15.9	5.67
3709	1.55	21.0	7.17	1.48	16.1	9.15
3703	1.57	19.3	8.76	1.65	16.3	10.1
3691	1.60	17.3	11.3	1.92	16.6	11.5
3689	1.64	14.4	14.0	1.98	17.0	11.6
3686	1.66	11.6	16.5	1.87	17.3	10.8
3688	1.67	9.02	18.5	1.64	17.4	9.44
3691	1.68	6.61	19.7	1.28	17.5	7.32



Performance characteristic curves of the KAIAO KDP40 self-priming centrifugal pumpset

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : KAIAO Centrifugal Pumpset
Model : KDP40

Date of Test : August 9, 2013

Test Condition

Ambient Temperature (°C) : 27.2
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 28.2

Table 3. Results of Cavitation Test

PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
3687	0.68	9.17	8.54	19.5
3692	1.64	8.21	9.37	19.3
3695	3.03	6.82	10.5	18.0
3704	4.45	5.40	11.7	15.9
3719	5.90	3.95	12.3	11.8
3735	6.62	3.23	12.7	9.13
3739	7.33	2.52	13.2	6.84
3743	8.05	1.81	13.7	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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DELFIN C. SUMINISTRADO
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N.B.

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